

NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

Date	07/07/26
Team ID	XXXXXX
Project Name	edugenie google gemini powered learning assistant
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib.
1	Interactive AI Tutor Chat	Real-time conversation interface utilizing Google Gemini API to explain complex educational concepts, solve equations, and answer student queries dynamically.	Gemini Integration / Chat Module	Done	1.0
2	Automated Quiz Generation	Generates personalized multiple-choice and short-answer quizzes based on any uploaded textbook chapter or lecture notes topic.	Assessment Engine	Done	1.0
3	Smart Study Material Summarizer	Accepts long PDFs/text documents and creates concise bulleted summaries, action items, and key flashcards using LLM parsing.	Content Processing	Done	1.0
4	Voice-Enabled Learning Mode	Allows students to interact with the learning assistant using speech-to-text and text-to-speech for accessible, hands-free learning.	Audio Interface	In Progress	0.5
5	Personalized Dashboard & Analytics	Tracks student progress, visualizes strengths/weaknesses across topics, and suggests customized daily learning pathways.	Analytics / UI	Done	1.0
6	Secure User Auth & Profile Storage	Enables users to safely sign up, log in, and securely save their learning history, bookmarks, and historical test scores.	Authentication / DB	Done	0.5

Feature Summary

Metric	Count/Value
Total Features Planned	6
Total Features Implemented	5
Core / Must-Have Features	4

Additional / Nice-to-Have Features

- **Gamified Badges & Rewards System:** Unlocks performance achievements and streaks to boost student engagement and motivation (Planned / Next Phase).
- **Multi-Language Support:** Leverages Gemini's multilingual capabilities to let users study and ask questions in native languages like Spanish, Hindi, and French.
- **Collaborative Study Rooms:** Allows peer-to-peer connection where multiple students can interact with the same EduGenie session.

Features Tested & Verified

- **Gemini API Prompt Reliability:** Rigorously verified structured JSON outputs from Gemini to prevent UI parsing failures during quiz generation.
- **Session Persistence:** Verified that conversation histories and user dashboard metrics are accurately stored and reloaded from the database.
- **Responsive UI Interface:** Tested cross-device compatibility across mobile, tablet, and desktop viewports to ensure seamless visual layout.

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Dashboard, Chat Window, Quiz Panel	Responsive Sidebar navigation, Interactive chat bubbles, Markdown rendering for math formulas.
2	Backend/Logic	AI Orchestrator, Prompt Router	Context handling window, Session tokens counter, Dynamic prompt templates builder.
3	Database / Storage	User Profiles, Progress DB	SQL/NoSQL schemas storing chat history, generated quizzes, and daily analytics parameters.
4	API / Integration	Google Gemini API, TTS Engine	Secure HTTPS fetch requests, streaming response handlers, voice synthesis integration.
5	Security / Authentication	Session Manager, Encryption	JWT based access control, secure environmental variable handling for API secret keys.